The origin of ferromagnetism and the Hubbard model

part 3: Hubbard model
introduction and some elementary results

Advanced Topics in Statistical Physics
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§basic setting

sites

$\frac{1}{2}$ $-\frac{1}{2}$

▷ lattice (finite set) $\Lambda \ni x, y, \dots$ spin $\sigma = \uparrow, \downarrow$

▷ basic fermion operators

$\hat{c}^\dagger_{x\sigma}$ creates an electron at x with spin σ

$\hat{c}_{x\sigma}$ annihilates an electron at x with spin σ

$$\{\hat{c}_{x\sigma}, \hat{c}_{y\tau}\} = \{\hat{c}^\dagger_{x\sigma}, \hat{c}^\dagger_{y\tau}\} = 0, \quad \{\hat{c}_{x\sigma}, \hat{c}^\dagger_{y\tau}\} = \delta_{xy}\delta_{\sigma\tau} \qquad (1)$$

with $\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}$ (2)

$$\to (\hat{c}_{x\sigma})^2 = (\hat{c}^\dagger_{x\sigma})^2 = 0 \qquad (3)$$

▷ number operator $\hat{n}_{x\sigma} = \hat{c}^\dagger_{x\sigma}\hat{c}_{x\sigma}$ (4)

you can show $(\hat{n}_{x\sigma})^2 = \hat{n}_{x\sigma}$ (5) $\to$ the eigenvalues of $\hat{n}_{x\sigma}$ are 0, 1

"vacuum" and the Hilbert space

$|\Phi_{vac}\rangle$ the state with no electrons (in the relevant orbits)

$$\langle\Phi_{vac}|\Phi_{vac}\rangle=1 \quad (1) \qquad \hat{c}_{x\sigma}|\Phi_{vac}\rangle=0 \text{ for any } x,\sigma \quad (2)$$

fix electron number N with $0\leq N\leq 2|\Lambda|$ = the number of sites in Λ (3)

$$x_1,\ldots,x_N\in\Lambda,\quad \sigma_1,\ldots,\sigma_N\in\{\uparrow,\downarrow\} \quad (4)$$

state in which N electrons occupy single particle states $(x_1,\sigma_1),\ldots,(x_N,\sigma_N)$

$$\hat{c}^\dagger_{x_1\sigma_1}\hat{c}^\dagger_{x_2\sigma_2}\cdots\hat{c}^\dagger_{x_N\sigma_N}|\Phi_{vac}\rangle \quad (5)$$

- vanishes if $(x_i,\sigma_i)=(x_j,\sigma_j)$ for $i\neq j$
- states are "overcounted" → next page

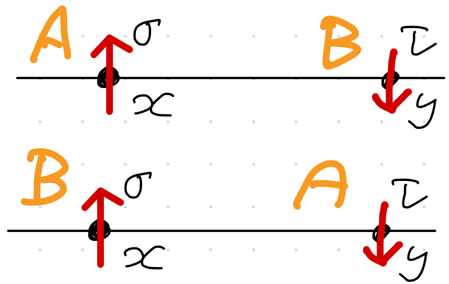
N-electron Hilbert space $\mathcal{H}_N$

spanned by all states of the form (5) ← $\dim\mathcal{H}_N<\infty$

states get a factor of -1 when we swap the order = labels

→ part 2 p. 2

$$\underbrace{\hat{c}^\dagger_{x\sigma}}_{A}\,\underbrace{\hat{c}^\dagger_{y\tau}}_{B}|\Phi_{\mathrm{vac}}\rangle = -\underbrace{\hat{c}^\dagger_{y\tau}}_{A}\,\underbrace{\hat{c}^\dagger_{x\sigma}}_{B}|\Phi_{\mathrm{vac}}\rangle \quad (1)$$



fermion symmetry is built into the operators!

▷ spin operators $\quad \hat{\mathbf{S}}_x = (\hat{S}^{(x)}_x, \hat{S}^{(y)}_x, \hat{S}^{(z)}_x) \qquad x \in \Lambda$

$$\hat{S}^{(z)}_x = \frac{1}{2}\{\hat{n}_{x\uparrow} - \hat{n}_{x\downarrow}\},\quad \hat{S}^{(x)}_x = \frac{1}{2}\{\hat{S}^+_x + \hat{S}^-_x\},\quad \hat{S}^{(y)}_x = \frac{1}{2i}\{\hat{S}^+_x - \hat{S}^-_x\} \quad (2)$$

$$\text{with } \hat{S}^+_x = \hat{c}^\dagger_{x\uparrow}\hat{c}_{x\downarrow},\quad \hat{S}^-_x = \hat{c}^\dagger_{x\downarrow}\hat{c}_{x\uparrow} \quad (3)$$

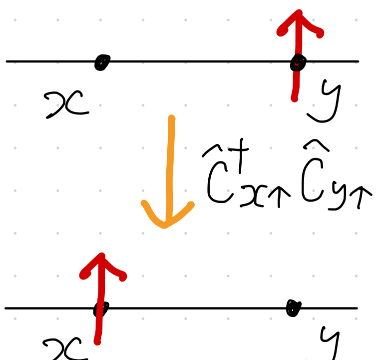
total spin operator $\hat{\mathbf{S}}_{\mathrm{tot}} = \sum_{x\in\Lambda} \hat{\mathbf{S}}_x$ (4) $\quad (\hat{\mathbf{S}}_{\mathrm{tot}})^2$ has eigenvalues $S_{\mathrm{tot}}(S_{\mathrm{tot}}+1)$

$$S_{\mathrm{tot}} = \begin{cases} 0, 1, \dots, \frac{N}{2}-1, \boxed{\frac{N}{2}} & (\text{even } N \le |\Lambda|) \\ \frac{1}{2}, \frac{3}{2}, \dots, \frac{N}{2}-1, \boxed{\frac{N}{2}} & (\text{odd } N \le |\Lambda|) \end{cases} \quad (5)$$

§ Hamiltonian of the Hubbard model

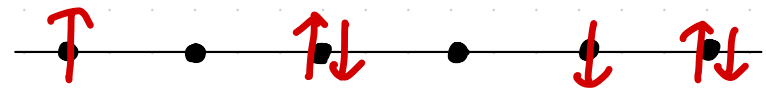
- hopping amplitude $t_{xy} = t_{yx} \in \mathbb{R}$ $\quad (x, y \in \Lambda)$ $\quad t_{xx} = V_x$
- on-site Coulomb energy $U > 0$

$$\hat{H} = \underbrace{\sum_{\substack{x,y\in\Lambda \\ \sigma=\uparrow,\downarrow}} t_{xy}\, \hat{c}^\dagger_{x\sigma} \hat{c}_{y\sigma}}_{\hat{H}_{\mathrm{hop}}} + \underbrace{U \sum_{x\in\Lambda} \hat{n}_{x\uparrow} \hat{n}_{x\downarrow}}_{\hat{H}_{\mathrm{int}}} \qquad (1)$$



electron hops from y to x

$U \times$ (the number of doubly occupied sites)



$2U$

▷ invariance under global spin rotation

$$[\hat{S}^{(\alpha)}_{tot}, \hat{H}] = 0 \quad (1) \qquad \alpha = x, y, z$$

one can study simultaneous eigenstates of $\hat{H}$, $\hat{S}^{(z)}_{tot}$, $(\hat{\mathbf{S}}_{tot})^2$

hopping part

for every $x, y \in \Lambda$

$$[\hat{S}^{(z)}_{tot}, \sum_{\sigma=\uparrow,\downarrow} \hat{c}^\dagger_{x\sigma}\hat{c}_{y\sigma}] = 0 \quad (2)$$

$$[\hat{S}^{\pm}_{tot}, \sum_{\sigma=\uparrow,\downarrow} \hat{c}^\dagger_{x\sigma}\hat{c}_{y\sigma}] = 0 \quad (3)$$

interaction part

$$\hat{n}_x = \hat{n}_{x\uparrow} + \hat{n}_{x\downarrow} \quad (4)$$

$$[\hat{S}^{(\alpha)}_{tot}, \hat{n}_x] = 0 \text{ for every } x \quad (5)$$

$$(\hat{n}_x)^2 = (\hat{n}_{x\uparrow} + \hat{n}_{x\downarrow})^2 = \hat{n}_{x\uparrow} + 2\hat{n}_{x\uparrow}\hat{n}_{x\downarrow} + \hat{n}_{x\downarrow}$$

$$= \hat{n}_x + 2\hat{n}_{x\uparrow}\hat{n}_{x\downarrow} \quad (6)$$

iii) action on a single electron state

$$|\psi\rangle = \sum_{x\in\Lambda} \psi_x \hat{c}^\dagger_{x\uparrow} |\Phi_{vac}\rangle \quad (1) \qquad \psi_x \in \mathbb{C}$$

$$\hat{H}|\psi\rangle = \sum_{x,y,z\in\Lambda} t_{xy} \psi_z \hat{c}^\dagger_{x\uparrow} \underbrace{\hat{c}_{y\uparrow} \hat{c}^\dagger_{z\uparrow}}_{\delta_{yz} - \hat{c}^\dagger \hat{c}} |\Phi_{vac}\rangle$$

$$= \sum_{x} \left(\sum_{y} t_{xy} \psi_y \right) \hat{c}^\dagger_x |\Phi_{vac}\rangle \quad (2)$$

thus

$$\hat{H}|\psi\rangle = E|\psi\rangle \quad (3)$$

$$\Updownarrow$$

$$\sum_{y\in\Lambda} t_{xy} \psi_y = E \psi_x \quad \text{for all } x \quad (4)$$

single-particle tight-binding Schrödinger equation

$\boxed{\text{part2 P.14-(1)}}$

§ the model with $U=0$ $\rightsquigarrow$ non-interacting electrons

$$\hat{H}=\hat{H}_{\mathrm{hop}}=\sum_{\substack{x,y\in\Lambda\\ \sigma=\uparrow,\downarrow}} t_{xy}\hat{c}^\dagger_{x\sigma}\hat{c}_{y\sigma} \qquad (1)$$

single-particle energy eigenstates

$$\sum_{y\in\Lambda} t_{xy}\psi^{(j)}_y=\epsilon_j\psi^{(j)}_x \quad \text{for all } x\in\Lambda \qquad (2)$$

$$j=1,\dots,|\Lambda|,\quad \epsilon_j\le\epsilon_{j+1},\quad \langle\psi^{(j)},\psi^{(k)}\rangle=\delta_{j,k} \qquad (3)$$

the corresponding creation operator $\boxed{\hat{a}^\dagger_{j\sigma}=\sum_{x\in\Lambda}\psi^{(j)}_x\hat{c}^\dagger_{x\sigma}}$ (4)

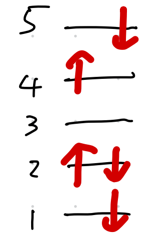
$$[\hat{H}_{\mathrm{hop}},\hat{a}^\dagger_{j\sigma}]=\sum_{\substack{x,y,z\\ \tau}} t_{xy}\psi^{(j)}_z\underbrace{[\hat{c}^\dagger_{x\tau}\hat{c}_{y\tau},\hat{c}^\dagger_{z\sigma}]}_{=\delta_{y,z}\delta_{\sigma,\tau}\hat{c}^\dagger_{x\tau}}=\sum_x\Big(\underbrace{\sum_y t_{xy}\psi^{(j)}_y}_{=\epsilon_j\psi^{(j)}_x}\Big)\hat{c}^\dagger_{x\sigma}$$

$$=\epsilon_j\hat{a}^\dagger_{j\sigma} \qquad (5)$$

$$\Big(\{\hat{a}_{j\sigma},\hat{a}_{k\tau}\}=\{\hat{a}^\dagger_{j\sigma},\hat{a}^\dagger_{k\tau}\}=0 \quad (6) \qquad \{\hat{a}_{j\sigma},\hat{a}^\dagger_{k\tau}\}=\delta_{jk}\delta_{\sigma\tau} \quad (7)\Big)$$

$$[\hat{H}_{hop}, \hat{a}^\dagger_{j\sigma}] = \epsilon_j \hat{a}^\dagger_{j\sigma} \quad (1) \qquad \hat{H}_{hop}|\Phi_{vac}\rangle = 0 \quad (2)$$

for $J_\uparrow, J_\downarrow \subset \{1,2,\ldots,|\Lambda|\}$, $|J_\uparrow| + |J_\downarrow| = N$



$J_\uparrow = \{2,4\}$

$J_\downarrow = \{1,2,5\}$

$$\hat{H}_{hop}\underbrace{\left(\prod_{j\in J_\uparrow}\hat{a}^\dagger_{j\uparrow}\right)\left(\prod_{j\in J_\downarrow}\hat{a}^\dagger_{j\downarrow}\right)|\Phi_{vac}\rangle}$$

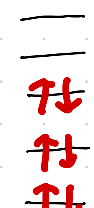
(Slater determinant state)

$$= \underbrace{\left(\sum_{j\in J_\uparrow}\epsilon_j + \sum_{j\in J_\downarrow}\epsilon_j\right)}_{\text{energy eigenvalue}}\underbrace{\left(\prod_{j\in J_\uparrow}\hat{a}^\dagger_{j\uparrow}\right)\left(\prod_{j\in J_\downarrow}\hat{a}^\dagger_{j\downarrow}\right)|\Phi_{vac}\rangle}_{\text{energy eigenstate}} \quad (3)$$

ground state for even N → (unique ground state if $\epsilon_{\frac{N}{2}} < \epsilon_{\frac{N}{2}+1}$)

$$|\Phi_{GS}\rangle = \left(\prod_{j=1}^{N/2}\hat{a}^\dagger_{j\uparrow}\hat{a}^\dagger_{j\downarrow}\right)|\Phi_{vac}\rangle \quad (4) \qquad \text{with } E_{GS} = 2\sum_{j=1}^{N/2}\epsilon_j \quad (5)$$

has $S_{tot} = 0$



$$\left(\hat{a}^\dagger_{j\uparrow}\hat{a}^\dagger_{\downarrow j} = \frac{1}{2}(a^\dagger_{j\uparrow}a^\dagger_{j\downarrow} - a^\dagger_{j\downarrow}a^\dagger_{j\uparrow}) \quad (6)\right)$$

Pauli paramagnetism

§ the model with $t_{xy}=0$ for all $x,y\in\Lambda$

$$\hat{H}=\hat{H}_{\rm int}=U\sum_{x\in\Lambda}\hat{n}_{x\uparrow}\hat{n}_{x\downarrow}$$

for $\Gamma_\uparrow,\Gamma_\downarrow\subset\Lambda$, $|\Gamma_\uparrow|+|\Gamma_\downarrow|=N$

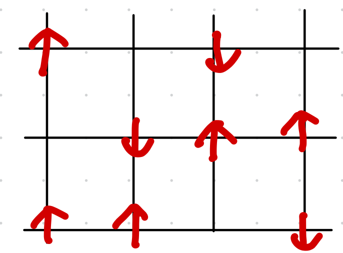
$$\hat{H}_{\rm int}\boxed{\left(\prod_{x\in\Gamma_\uparrow}\hat{c}^\dagger_{x\uparrow}\right)\left(\prod_{x\in\Gamma_\downarrow}\hat{c}^\dagger_{x\downarrow}\right)|\Phi_{\rm vac}\rangle}$$

same as the basis state p2-(5)
$\hat{c}^\dagger_{x_1\sigma_1}\cdots\hat{c}^\dagger_{x_N\sigma_N}|\Phi_{\rm vac}\rangle$

$$=\underbrace{U|\Gamma_\uparrow\cap\Gamma_\downarrow|}_{\text{energy eigenvalue}}\left(\prod_{x\in\Gamma_\uparrow}\hat{c}^\dagger_{x\uparrow}\right)\left(\prod_{x\in\Gamma_\downarrow}\hat{c}^\dagger_{x\downarrow}\right)|\Phi_{\rm vac}\rangle$$

any state with mimium $|\Gamma_\uparrow\cap\Gamma_\downarrow|$ is a ground state

for $N\leq|\Lambda|$, $\Gamma_\uparrow\cap\Gamma_\downarrow=\phi$ with $E_{GS}=0$



highly degenerate ground states → no magnetic order $\boxed{\text{paramagnetism}}$

§ the Hubbard model

$$\hat{H} = \boxed{\hat{H}_{hop}} + \boxed{\hat{H}_{int}} \qquad (1)$$

wave-like nature of electrons

particle-like nature of electrons

favors Pauli paramagnetism

wave-particle dualism !!

favors trivial paramagnetism

"competition" between $\hat{H}_{hop}$ and $\hat{H}_{int}$ may lead to nontrivial physics including ferromagnetism !

§ saturated ferromagnetism for $N \le |\Lambda|$

definitions

saturated ferromagnetism at zero temperature

- a Hubbard model exhibits ferromagnetism if any ground state has $S_{tot} = N/2$ ← maximum possible S_{tot}
- we say "ground state is unique" if ground states have only trivial $2S_{tot}+1$ fold degeneracy

simplification

the sector with $S^{(z)}_{tot} = S_{tot} = N/2$ → only ↑ spins

Hamiltonian is essentially $\hat{H}_{\uparrow} = \sum_{x,y\in\Lambda} t_{xy}\, \hat{c}^{\dagger}_{x\uparrow} \hat{c}_{y\uparrow}$ (1)

non-interacting fermions

ground state

$$|\Phi_{GS}^{S^{(z)}_{tot}=\frac{N}{2}}\rangle = \left(\prod_{j=1}^{N} \hat{a}^{\dagger}_{j\uparrow}\right)|\Phi_{vac}\rangle \quad (2)$$

Slater determinant

unique if $\epsilon_N < \epsilon_{N+1}$

$|\Phi_{GS}^{S_{tot}^{(z)}=M}\rangle$ a ground state with $S_{tot}^{(z)} = M$

$\left(M = -\frac{N}{2}, -\frac{N}{2}+1, \dots, \frac{N}{2}-1, \frac{N}{2}\right)$

$$|\tilde{\Phi}\rangle = \left(\hat{S}_{tot}^{+}\right)^{\frac{N}{2}-M} |\Phi_{GS}^{S_{tot}^{(z)}=M}\rangle \quad (1)$$ is a ground state with $S_{tot}^{(z)} = \frac{N}{2}$

all up!

and

$$|\Phi_{GS}^{S_{tot}^{(z)}=M}\rangle = \text{const.}\left(\hat{S}_{tot}^{-}\right)^{\frac{N}{2}-M} |\tilde{\Phi}\rangle \quad (2)$$

(because $$\left(\hat{S}_{tot}^{-}\right)^{n}\left(\hat{S}_{tot}^{+}\right)^{n} = \prod_{r=0}^{n-1}\left\{\left(\hat{\mathbf{S}}_{tot}\right)^2 - \left(\hat{S}_{tot}^{(z)}+r\right)\left(\hat{S}_{tot}^{(z)}+r+1\right)\right\} \quad (3)$$)

$\left(\hat{\mathbf{S}}_{tot}\right)^2 \to \frac{N}{2}\left(\frac{N}{2}+1\right)$

it suffices to understand ground states with $S_{tot}^{(z)} = \frac{N}{2}$

once we know there is saturated ferromagnetism

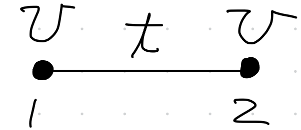
NONTRIVIAL!!

if $E_N < E_{N+1}$

$$|\Phi_{GS}^{S_{tot}^{(z)}=M}\rangle = \text{const.}\left(\hat{S}_{tot}^{-}\right)^{\frac{N}{2}-M}\left(\prod_{j=1}^{N}\hat{a}_{j\uparrow}^{\dagger}\right)|\Phi_{vac}\rangle \quad (4)$$

§ models with two electrons $N=2$

▷ two-site model $\Lambda=\{1,2\}$



$$\hat{H}=t\sum_{\sigma=\uparrow,\downarrow}\{\hat{c}^\dagger_{1\sigma}\hat{c}_{2\sigma}+\hat{c}^\dagger_{2\sigma}\hat{c}_{1\sigma}\}+U\{\hat{n}_{1\uparrow}\hat{n}_{1\downarrow}+\hat{n}_{2\uparrow}\hat{n}_{2\downarrow}\} \quad (1)$$

$$\left.\begin{aligned}&\hat{H}\,\hat{c}^\dagger_{1\uparrow}\hat{c}^\dagger_{2\uparrow}|\Phi_{vac}\rangle=0,\quad \hat{H}\,\hat{c}^\dagger_{1\downarrow}\hat{c}^\dagger_{2\downarrow}|\Phi_{vac}\rangle=0 \quad (2)\\ &\hat{H}\,\frac{1}{\sqrt{2}}(\hat{c}^\dagger_{1\uparrow}\hat{c}^\dagger_{2\downarrow}+\hat{c}^\dagger_{1\downarrow}\hat{c}^\dagger_{2\uparrow})|\Phi_{vac}\rangle=0 \quad (3)\end{aligned}\right\}\ \text{triplet } S_{tot}=1$$

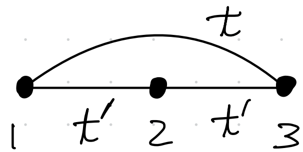
the ground state

$$|\Phi_{GS}\rangle=\{\alpha(\hat{c}^\dagger_{1\uparrow}\hat{c}^\dagger_{2\downarrow}-\hat{c}^\dagger_{1\downarrow}\hat{c}^\dagger_{2\uparrow})+\beta(\hat{c}^\dagger_{1\uparrow}\hat{c}^\dagger_{1\downarrow}+\hat{c}^\dagger_{2\uparrow}\hat{c}^\dagger_{2\downarrow})\}|\Phi_{vac}\rangle \quad (4)$$

with $E_{GS}=\frac{1}{2}\{U-\sqrt{U^2+16t^2}\}<0,\quad \alpha=2t\beta/E_{GS}$ (5)

has $S_{tot}=0$ consistent with Wigner's theorem (part2-p.10)

three-site model $\Lambda = \{1, 2, 3\}$



$$\begin{cases} t_{12} = t_{23} = \boxed{t'} > 0 \;\rightarrow\; \text{fix} \\ t_{13} = t \\ t_{jj} = 0 \end{cases}$$

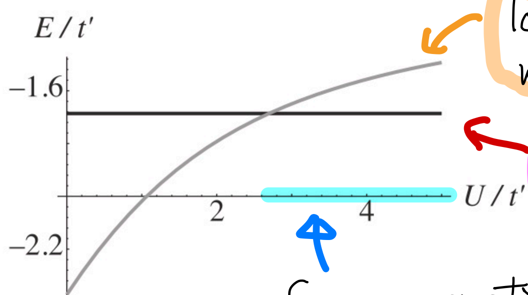
$U = \infty$ (part 2 - p.21)

$t < 0$ the ground state has $S_{tot} = 0$ → antiferromagnetism

$t > 0$ the ground state has $S_{tot} = 1$ → ferromagnetism!

$0 < U < \infty$

general case $t = \frac{t'}{2} > 0$

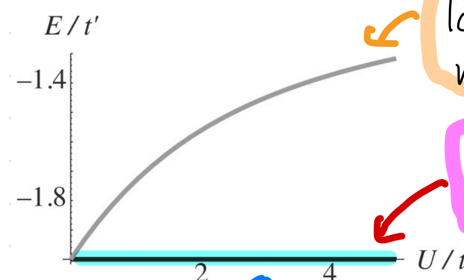


lowest energy with $S_{tot} = 0$

lowest energy with $S_{tot} = 1$

ferromagnetism!

special case $t = t' > 0$



lowest energy with $S_{tot} = 0$

lowest energy with $S_{tot} = 1$

ferromagnetism!!

§ Stoner criterion $\rightarrow$ non-rigorous mean-field argument

suppose there are $N_\uparrow$ up-spin electrons and $N_\downarrow$ down-spin electrons

▷ the minimizer of $\hat{H}_{hop}$

$$|\Phi^{N_\uparrow,N_\downarrow}_{MF}\rangle = \left(\prod_{j=1}^{N_\uparrow} \hat{a}^\dagger_{j\uparrow}\right)\left(\prod_{j=1}^{N_\uparrow} \hat{a}^\dagger_{j\downarrow}\right)|\Phi_{vac}\rangle \quad (1)$$

with energy $$\sum_{j=1}^{N_\uparrow}\epsilon_j + \sum_{j=1}^{N_\downarrow}\epsilon_j \quad (2)$$

▷ mean-field treatment of $\hat{H}_{int}$

$$\hat{H}_{int} = U\sum_{x\in\Lambda}\hat{n}_{x\uparrow}\hat{n}_{x\downarrow} \quad (3)$$

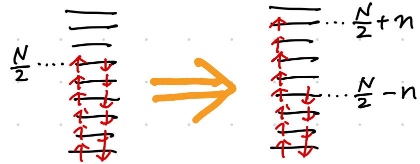
$$\rightarrow U\sum_{x\in\Lambda}\langle\hat{n}_{x\uparrow}\rangle\langle\hat{n}_{x\downarrow}\rangle = U\sum_{x\in\Lambda}\frac{N_\uparrow}{|\Lambda|}\frac{N_\downarrow}{|\Lambda|} = U\frac{N_\uparrow N_\downarrow}{|\Lambda|} \quad (4)$$

▷ mean-field energy

$$E_{MF}(N_\uparrow, N_\downarrow) = \sum_{j=1}^{N_\uparrow}\epsilon_j + \sum_{j=1}^{N_\downarrow}\epsilon_j + \frac{U}{|\Lambda|}N_\uparrow N_\downarrow \quad (5)$$

we examine the stability of the Pauli paramagnetic state $|\Phi_{MF}^{N/2,N/2}\rangle$

$$|\Phi_{MF}^{\frac{N}{2},\frac{N}{2}}\rangle$$



$$|\Phi_{MF}^{\frac{N}{2}+n,\frac{N}{2}-n}\rangle$$

$$n \ll N$$

$$\Delta E(n) = E_{MF}\left(\frac{N}{2}+n,\frac{N}{2}-n\right) - E_{MF}\left(\frac{N}{2},\frac{N}{2}\right)$$

$$= \sum_{j=1}^{\frac{N}{2}+n}\epsilon_j + \sum_{j=1}^{\frac{N}{2}-n}\epsilon_j + \frac{U}{|\Lambda|}\left(\frac{N}{2}+n\right)\left(\frac{N}{2}-n\right) - \left\{\sum_{j=1}^{\frac{N}{2}}\epsilon_j + \sum_{j=1}^{\frac{N}{2}}\epsilon_j + \frac{U}{|\Lambda|}\frac{N^2}{4}\right\}$$

$$= \sum_{l=1}^{n}\left(\epsilon_{\frac{N}{2}+l} - \epsilon_{\frac{N}{2}-n+l}\right) - U\frac{n^2}{|\Lambda|} \simeq \sum_{l=1}^{n} n\Delta\epsilon - U\frac{n^2}{|\Lambda|}$$

$\Delta\epsilon$ → mean level spacing

$$\simeq \frac{n^2}{|\Lambda|D_F} - U\frac{n^2}{|\Lambda|} = \frac{n^2}{|\Lambda|D_F}(1-D_F U) \quad (1)$$

$$\Delta\epsilon \simeq \frac{1}{|\Lambda|D_F} \quad (2)$$

D_F: single-particle density of states per volume

$\Delta E(n) < 0$ (paramagnetism is unstable) when

$$\boxed{D_F U > 1} \quad (3)$$

Stoner criterion for ferromagnetism